

Literature status: total incomparability and singularity

Source arXiv:1908.09228; answer arXiv:1906.06677

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Status: literature already answered (negative).

Original problem

In Problem 8.10 of [1] (source PDF p. 20), after Propositions 8.8 and 8.9, the authors ask: if X_θ is totally incomparable with X_t for every $t \neq \theta$ in a neighborhood of θ , must the associated interpolation differential Ω_θ be singular? The next sentence explicitly points to [2] for the negative solution.

Supporting counterexample

Section 8.1 of [2] (supporting PDF p. 35), titled “A totally incomparable family with nonsingular derivation at any point”, gives the promised construction. Starting from the couple (ℓ_∞, ℓ_1) and its second Rochberg spaces, the authors obtain a family whose spaces at distinct real parameters are pairwise totally incomparable (by $\ell_{1/t}$ -saturation), whereas the associated differentials are not singular at any parameter. This violates the proposed implication exactly.

The supporting authors explicitly know they are answering this problem; the relation is not an agent-derived reformulation. The existing `1906.06677_rochberg_E.mn_embedding` packet concerns a separate question and is unaffected.

Search evidence

The original paper names arXiv:1906.06677 in the sentence announcing the negative solution. Inspection of that source located the explicit proposition and proof in its applications/counterexamples section.

References

- [1] J. M. F. Castillo, W. H. G. Corrêa, and F. Cabello Sánchez, *Differential processes generated by two interpolators*, arXiv:1908.09228 (2019).
- [2] J. M. F. Castillo, W. H. G. Corrêa, and F. Cabello Sánchez, *Higher order derivatives of analytic families of Banach spaces*, arXiv:1906.06677 (2019).